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Legal Assist AI-Leveraging Transformer based Model for Effective Legal Assistance

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Abstract

Pursuit of accessible legal assistance in India faces a critical gap, as many citizens struggle to leverage their legal rights due to limited awareness and access to relevant legal information. This paper introduces Legal Assist AI, a transformer-based model designed to bridge this gap by offering effective legal assistance through large language models (LLMs). The system leverages advanced natural language processing (NLP) techniques to analyze and respond to both subjective and objective legal queries, catering to diverse users such as legal professionals, scholars, and the general public. The model was fine-tuned on extensive datasets from the Indian legal domain, including the *Lawyer_GPT_India* and All India Bar Examination (AIBE) dataset, providing a robust understanding of the complexities of Indian law. The model was evaluated against state-of-the-art models such as GPT-3.5 Turbo and Mistral 7B, achieving a 60.08% score on the AIBE, outperforming its competitors in legal reasoning and accuracy. Unlike other models, Legal Assist AI avoided common issues such as hallucinations, making it highly reliable for practical legal applications. It showcases the model's applicability in real-world legal scenarios, with future iterations aiming to enhance performance and expanding its dataset to cover a broader range of multilingual, and case-specific queries as well.

1. Introduction

Justice is a fundamental right in India protected by Articles 14 and 21 of the Indian Constitution [1]. The Right to Information (RTI) Act 2005 [2] allows citizens to seek knowledge and transparency with respect to legal proceedings. Despite these acts and articles enshrined in the Constitution of India, there is still a significant gap between the legal rights and the public's awareness of this information. This prevents many people from taking the legal recourse they would have taken had they had easy access to adequate legal information [3]. This gap or disparity affects different groups differently, viz., the general public seeks this information to understand their rights, scholars aim to conduct in-depth research, and professionals need to fact-check information. Access to legal resources is thus crucial for ensuring that justice is not just a theoretical entity but could be exercised as a practical reality for all. In this context, one of the potential areas of change is the implementation of large language models (LLMs) in the legal system. Harnessing the power of natural language processing (NLP) with deep learning, a machine learning powered legal assistant can understand and generate human-like text for legal queries [4]. Transformer-based legal models could aid in optimizing processes and democratizing access to legal resources for the general population [5, 6]. LLMs significantly reduce operational costs compared to traditional approaches, achieving an impressive 99.97 percent savings. The use of LLMs enable legal departments and law firms to significantly scale their contract review throughput without a proportional increase in costs [7]. This paper proposes “Legal Assist” which in concept and application is a robust legal assistant. The proposed model has been tested for two different unstructured lines of questioning, i.e. long form (subjective) and objective (four options).

2. Literature Review

A study dependent on ChatGPT (GPT-3.5 Turbo), generated legal responses in plain English, avoiding sophisticated legal terms, yet at times resulted in verbose and repetitive responses. One of the biggest drawbacks of ChatGPT is that it hallucinates (incorrect information generation), which sometimes confuses the user, even though the scalability of the platform is good and it helps to generate any type of information [8, 9].

A study conducted by Tiwari et al. [10] investigated the application of GPT-3.5 Turbo as a base model for legal querying. The model was tested on only a few sets of data to check its capability in the legal and paralegal domains, but it has come out to be only model so far to qualify the All-India Bar Examination (AIBE). The same study explored Mistral 7B as the base model and was tested on multiple parameters and datasets. This model performed quite well in resolving the core problem or legal question that needed prompt addressing. But it exhibited significant lag in performance in comparison to GPT-3.5-Turbo, as it scored low in Legal Bench Metrics and couldn't pass the AIBE exam. The paper also discussed the AALAP (AI Assistant for Legal and Paralegal Functions in India) model, which is a fine-tuned Mistral 7B trained on the AALAP dataset. The performance of AALAP was found to be better than 85% of test data for Mistral 7B, better than 31% of the test data for GPT-3.5 Turbo, and AALAP achieved an equivalent score in 34% of the test data evaluated by GPT-4. AALAP did poorly in the AIBE exam. The model also did not manage to score well in the LegalBench metrics as compared to GPT-3.5 Turbo [10].

A paper presents Paramanu, a family of Generative Large Language models designed to efficiently handle legal language processing in 10 Indian languages across 5 scripts. These models outperformed larger counterparts in legal text analysis for Bangla, Hindi, and Sanskrit, making them accessible and efficient for legal applications without requiring high computational power [11].

Evaluation of the law-specific LLMs has to be done using benchmarks that would stress legal tasks. Because the legal processes of the different countries vary so significantly, it is very difficult to create a single global evaluation benchmark for the legal LLM. LegalBench is a collaboratively built legal reasoning benchmark, consisting of 162 tasks in the USA law context [12]. LawBench [13] is a similar benchmark in the context of China. The following Table 1 presents a comparative overview of existing model architectures discussed above.

Table 1
Review of previous years' relevant Works/Models

Paper/Year	Model	Parameters	Dataset	Test Jurisdiction	Results
[9] **2023**	ChatGPT (3.5 Turbo)	175 billion	-	Canadian Laws	Good at providing more generalised legal information Often hallucinates (incorrect information generation) Often lacks accuracy
	JusticeBot	-	-	Canadian Law	Higher level of completeness in providing legal information Less generalised
[10] **2024**	Mistral 7B	7 billion	-	Indian Law	Failed the AIBE scoring 23.48%
	AALAP	7 billion	AALAP	Indian Law	Failed the AIBE scoring 25.56% Performed better on AALAP dataset as compared to GPT-3.5 Turbo
	GPT-3.5 Turbo	175 billion	-	Indian Law	Passed the AIBE test scoring 58.72% Performed better on LegalBench Dataset
[11] **2024**	Parmanu	-	The constitution of India, IPC and Case documents from the Supreme Court of India	Indian Law	Evaluated using clarity, relevance, completeness, and legal reasoning and achieved scores above 8/10 in all categories. No measurement of hallucination. Potential for generating biased or factually incorrect information due to high creativity settings during text generation.

3. Methodology

3.1. Data Preparation

Legal documents pertaining to the Indian Judicial System, including the Indian Constitution, Bharatiya Nyaya Sanhita (BNS), Bharatiya Nagarik Suraksha Sanhita (BNSS), Bharatiya Sakshya Adhiniyam (BSA), Labor Laws, State Legislatures, SEBI Guidelines, Judicial Judgements, AIBE (All India Bar Examination) Syllabus, and so forth were used in this study. Multiple legal contracts, their formats, etc., were also included in the dataset. All the legal documents for the study were downloaded from governmental and public websites, including court decisions, legislative text, and legal guidelines.

The dataset was then loaded using Directory Loader in the Langchain framework. This utility was used to effectively handle different tasks in natural language processing by aggregating and combining datasets within several directories. Langchain provides a strong support system for all such operations, ensuring that the data is prepared correctly and incorporated with ease into the model's training pipeline.

Further, the documents were broken into manageable chunks. A chunk size of 1000 characters with an overlap of 20 characters across adjacent chunks was preferred to maintain continuity of context between chunks, which was important in maintaining fluency during legal document processing. This made the process of segmentation optimal and implied an improved treatment of long and complex legal texts.

Embeddings are dense vector representations of text that encode its semantic meanings, thus making it possible to treat much larger models in a better way and understand languages more. These vectors transform text into a numerical format that can be utilized for classification and similarity searches. It uses embeddings to compare and analyze content in terms of meaning. For generating the embeddings of each text chunk, HuggingFaceEmbeddings was used. The sentence-transformers/all-MiniLM-L6-v2 model was chosen for its effective and quick embeddings. This allowed a powerful base in indexing and retrieval for a legal assistant model.

After generating the embeddings, they were indexed in a process that uses FAISS (Facebook AI Similarity Search), a high-speed library for similar vector search and clustering. FAISS indexes made it possible to conduct fast and scalable searches by indexing the embeddings into a vector store that had been optimized for the retrieval of similar vectors at speed. This is important for comparing and finding similar text chunks, which many other processes in legal document analysis required.

After indexing, the vector store was saved locally, and that was easier for access in further queries and analysis. The embeddings were kept in a local vector store to allow effective information retrieval by boosting performance and accuracy in legal document processing.

3.2. Proposed Framework

3.2.1. Model Selection and Architecture

Table 2 given below compares different large language models across a variety of tasks and benchmarks where Llama 3.1 8B is found to be the best performing across majority of benchmarks. Therefore, Llama 3.1 8B [14], was selected as the base model for this study. The model is characterized by high performance in multilingual tasks, a maximum context length of 128K tokens, and enhanced reasoning capabilities. With 8.03 billion parameters, the model has an optimum balance between computational efficiency and accuracy that suits the specific application needs of this research. [15].

Table 2
Comparison of Transformer Models [15]

Category	Benchmark	Llama 3.1 8B	Gemma 2 9B IT	Mistral 7B Instruct	GPT 3.5 Turbo 175B
General	**MMLU** (0-shot, CoT)	**73.0**	72.3 (5-shot, non-CoT)	60.5	69.8
	MMLU PRO (5-shot, CoT)	48.3	-	36.9	**49.2**
	IFEval	**80.4**	73.6	57.6	69.9
Code	**HumanEval** (0-shot)	**72.6**	54.3	40.2	68.0
	MBPP EvalPlus (base) (0-shot)	72.8	71.7	49.5	**82.0**
Math	**GSM8K** (8-shot, CoT)	**84.5**	76.7	53.2	81.6
	MATH (0-shot, CoT)	**51.9**	44.3	13	43.1
Reasoning	**ARC Challenge** (0-shot)	83.4	87.6	74.2	83.7
	GPQA (0-shot, CoT)	**32.8**	-	28.8	30.8
Tool use	**BFCL**	76.1	-	60.4	**85.9**
	Nexus	**38.5**	30	24.7	37.2
Long context	**ZeroSCROLLS/QuALITY**	81	-	-	-
	InfiniteBench/En.MC	65.1	-	-	-
	NIH/Multi-needle	98.8	-	-	-
Multilingual	**Multilingual MGSM** (0-shot)	**68.9**	53.2	29.9	51.4

The Transformer-based model, shown in Fig. 1, integrates Grouped Query Attention (GQA), which enhances memory efficiency during autoregressive decoding by using a single key-value head for multiple queries. Rotary Positional Embeddings (RoPE) are employed to capture the relative positional information between tokens, improving token interaction. The model utilizes RMS normalization before the self-attention and feedforward blocks to stabilize gradients and enhance training efficiency. Additionally, the

SwiGLU activation function is used in the feedforward layers, combining the Swish function to boost performance [16].

The Proposed Legal Assist AI Architecture, shown in Fig. 2, integrates a Retrieval QA Chain and a Large Language Model (LLM) to efficiently handle legal queries. The model processes input text tokens (queries) by retrieving relevant information from a vectorstore using semantic search, which fetches contextual data to enhance the query. This enriched query is then passed to the LLM, which generates a legal response based on the query and its context. The response is finally converted into output text tokens. By combining retrieval-based techniques with LLM capabilities, this architecture ensures accurate and context-aware answers to complex legal questions.

The given function $f(x)$ represents the process of generating a response based on an input query x. The process involves performing a semantic search to retrieve relevant contextual data $S(x)$, which is then combined with a predefined prompt P and the original query. This combined input is processed by a large language model L to produce the final response y.

$$\varvec{f}(\varvec{x}) = \varvec{L}(\varvec{P} + \varvec{S}(\varvec{x}) + \varvec{x}) = \varvec{y}$$

where:

- $\varvec{x}$: Input tokens (query)
- $\varvec{S}(\varvec{x})$: Contextual data retrieved from the semantic search using the input tokens
- $\varvec{P}$: Prompt
- $\varvec{L}$: Large language model that processes the combined input
- $\varvec{y}$: Generated response

3.2.2. Training

The quantized version of the transformer model had been used, with Q4_0 as the quantization method. The methodology had brought down the model size to circa 4.7 GB, which had significantly improved the inference time and efficiency with no loss in high accuracy. The choice of the type of quantization to apply had been informed by the necessity to be able to run this model in a very resource-constrained environment without sacrificing performance. [14]

The original base model had been pretrained on more than 15 trillion tokens from publicly available sources. Later, the prepared set of embedded vector store had been fine-tuned on the model by adjusting the parameters of the model. This process had focused on maximizing the model's performance in legal question answering while making use of the detailed nuances and complex structures in legal texts to ensure the model's understanding and responses to a broad spectrum of legal questions had been accurate and relevant.

4. Model Evaluations

To evaluate the effectiveness and robustness of the legal model, testing had been done with two sets of prompts over two separate datasets. Each type of prompt had been such that it had the ability to put pressure on different looks of the model, hence helping in getting a more complete set of estimations for the performance of the model. The following datasets had been used:

1. Lawyer_GPT_dataset [17]

2. AIBE dataset [18]

4.1. Evaluation using Lawyer_GPT_India dataset

This test had been conducted to understand the model's performance based on subjective reasoning. A dataset with different kinds of subjective questions had been chosen. The BERT score had also been calculated to see the model's understanding and its paraphrasing power and to check whether had generated the answer in the right context or not.

4.1.1. Dataset:

The dataset used here was sourced from HuggingFace. It has 150 pairs of questions within the domain of Indian Polity. The aim is to generate answers to a very wide range of questions related to the Indian Constitution, its judiciary, the legislative, and other prevailing socio-political issues of the country. This will be quite handy for a student/researcher or AI systems intending to understand or provide responses to questions on Indian Polity.

4.1.2. Data Fields:

1. Question: A text field that holds the question.

2. Answer: A text field that holds the matching answer.

4.1.3. Prompt:

The given prompt was used for generating subjective answers for the 150 questions.

System:

You are a respectful and honest legal assistant lawyer about Indian law. Answer under 50 words. If you don't know the answer, just say you don't know. Don't try to make up an answer.

Context:

{context}

User:

{question}

Response:

4.1.4 Observations:

i. The model had been iterated to answer all 150 questions with the given prompt template. Then each generated answer and actual answer was checked for semantic similarity using BERT Score. A sample of generated responses is given in Table 3 below.

Table 3: Sample of Generated responses for questions asked over Lawyer_GPT_dataset

question	answer	generated_answer	BERT score
“What is the key issue in the legal scenario: 'In Mehta v. Kapoor, Mehta, a tenant, sues Kapoor, a landlord, under the Indian Rent Control Act, for failure to provide basic amenities in the rented property. Mehta faced a blackout due to faulty wiring, which Kapoor knew about.'?”	“The key issue in Mehta v. Kapoor is whether the landlord (Kapoor) is liable for the harm to the tenant (Mehta) due to negligence in providing basic amenities, specifically, maintaining the electrical wiring.”	“The key issue in this legal scenario is the landlord's (Kapoor) liability for providing basic amenities and safety measures in the rented property, specifically addressing the tenant's (Mehta) complaint of a blackout caused by faulty wiring.”	88%
“Can you summarize the main arguments of the Supreme Court of India judgment in Kesavananda Bharati v. State of Kerala?”	“In Kesavananda Bharati v. State of Kerala, the Supreme Court ruled that the basic structure of the Constitution, including inviolable fundamental rights, cannot be altered by the Parliament.”	“The Supreme Court ruled by a 7-6 majority that the legislative power to amend the Constitution under Article 368 is subject to judicial review based on the doctrine of the "basic structure" of the Constitution. This means that Parliament cannot amend certain fundamental rights, such as property rights, in a way that alters the basic structure of the Constitution.”	80%
“Can you explain the legal term 'Public Interest Litigation (PIL)' in simple language?”	“Public Interest Litigation (PIL) is a legal action initiated in a court of law for the enforcement of public interest or general public welfare. It allows any member of the public to file a lawsuit, not just someone directly affected by the issue.”	“In simple terms, Public Interest Litigation (PIL) is a way for citizens or organizations to file cases on behalf of victims who have been harmed by government actions or policies that are against the law. It's like a "public complaint" system where people can bring attention to injustices and seek justice on behalf of others.”	76%
“What is the meaning of the following provision in the Indian Penal Code: 'Whoever, intending to take dishonestly any movable property out of the possession of any person without that person’s consent, moves that	“This provision describes theft. If a person intends to dishonestly take movable property from someone without their consent, and acts on this by moving the property, they are committing theft.”	“The provision states that a person commits theft if they intend to take something (movable property) from someone else's possession without their consent by moving it. This means that the act of taking or moving the property with the intention of stealing it constitutes theft.”	97%

property, is said to commit theft.'?”			
“Can you write an opening statement for a defence attorney in a defamation case under Indian law?”	“Honourable court, my client has been unjustly accused of defamation. However, as the evidence will show, my client's actions were either truthful or represent fair comments made in the public interest. I urge you to remember that, under our law, a statement is not defamatory if it can be proven to be true or an opinion.”	“Here's a possible opening statement: "Ladies and gentlemen of the jury, esteemed members of the court, today we gather to examine a claim of defamation made against my client. As we delve into this matter, I urge you to consider the nuances of free speech and the importance of protecting it under Indian law. The plaintiff alleges that my client's words were malicious and caused harm to their reputation. However, as we will demonstrate through evidence and testimony, my client's statements were made in good faith and were substantially true. We will show that the plaintiff's own actions and conduct have been subject to scrutiny and criticism by others, and that my client's comments were merely a response to those criticisms. We will also present expert testimony on the standards of public discourse and the limits of free speech under Indian law. I ask you to consider whether the plaintiff's reputation has truly been harmed by my client's words, or if this is simply an attempt to silence legitimate criticism. I urge you to carefully weigh the evidence and apply the principles of Indian law as we navigate this complex case.””	84%

ii. This test also revealed that the model proposed here was not hallucinating. When asked a question that the model did not know, the model replied, "I don't know." For Example:

Question: “What are the legal implications of data localization requirements under the Personal Data Protection Bill?”
Response: “I don't know. The provided text does not mention data localization requirements, and I'm unable to provide information on this topic based on the given context.
Question: “Explain the idea of a 'National Climate Court' and its potential role in India.”
Response: “I don't know. The provided text does not mention the concept of a "National Climate Court" or its potential role in India. It discusses India's international commitments, obligations towards preventing climate change, and the need for holistic consideration of competing considerations, but it does not introduce this specific idea.”

In contrast, GPT-3.5 powered ChatGPT often hallucinated providing inaccurate or fabricated information [8, 9]. Table 4 summarizes the GPT 3.5 Turbo and Legal Assist AI on basis of hallucinations.

Table 4: Comparison of Models on the Basis of Hallucinations

Metrics	GPT-3.5 Turbo	Legal Assist AI
Hallucinations	Hallucinates	No Hallucinations

4.2 Evaluation using All India Bar Examination dataset

This test was conducted to assess the model's knowledge domain and to evaluate its accuracy based on objective-based questions. By leveraging a standardized legal examination, the evaluation aimed to determine how effectively the model could understand and respond to various legal scenarios.

4.2.1. Dataset

All India Bar Examination is a test conducted by the Bar Council of India for law graduates who are interested in practicing as advocates. The candidate will be provided with a certificate of practice on passing the examination and can appear in any court throughout India. The dataset is made up of questions that have been drawn from the AIBE examinations running from AIBE 4 to AIBE 16, and it covers a time span of 12 years. There are a total of 1156 questions, all multiple-choice. The passing grade in the examination stands at 40%.

4.2.2. Data Fields

1. exam_name
2. exam_number
3. question_number
4. question_text
5. options
6. Answer

4.2.3. Prompt:

The given prompt was used for generating one-letter objective-based answers for the AIBE question dataset.

System:

You are a respectful and honest legal assistant lawyer about Indian law; just answer the correct option as either A, B, C, or D only; don't print other options. \ Don't give unnecessary text or change the option sequence.

Context:

{context}

User:

{question}

Response:

4.2.4. Observations:

The model had been tested on the given prompt template to generate one-letter responses from the given options. Out of 1156 questions, 682 were correct, resulting in an accuracy of 58.94555%.

However, there were 22 questions for which the answers provided differed from the options specified. These questions were manually reviewed by experts, who found that the options' answers were no longer relevant. Conversely, manual verification of the generated answers revealed that the model's predictions for these 22 questions were accurate and up-to-date. To prevent skewing the outcome, the authors decided to exclude these 22 questions. Consequently, the Legal Assist AI model's accuracy was found to be 60.08%. Table 5 presents a comparative analysis of multiple models based on the AIBE score.

Table 5: Comparison of Models on the Basis of AIBE Score

Metrics	Mistral 7B	AALAP	GPT-3.5 Turbo	Legal Assist AI
AIBE Score	23.48%	25.56%	58.72%	**60.08%**
AIBE Results	Fail	Fail	Passed	Passed

5. Discussions

Previous legal assistance models, such as GPT3.5 Turbo [19], Mistral 7B [20], and AALAP, have struggled to achieve better performance due to limitations in their training datasets. The scarcity of high-quality, domain-specific, and case specific training data hindered the development of effective legal AI models. The proposed Legal Assist AI Model outperformed the current state-of-the-art models such as GPT3.5 Turbo, Mistral 7B, and AALAP [4] in delivering results and insights. The proposed Legal Assist AI model achieved a score of 60.08% on the AIBE examination, outperforming all other models. Notably, unlike GPT 3.5 Turbo, the proposed model did not hallucinate. The superior performance of the Legal Assist AI Model highlights the importance of high-quality training data in developing effective legal AI solutions. However, due to the lack of data specific to the case, Legal Assist AI model was unable to answer few questions. Still, this can be further generalized by enhancing the training corpus and adding more case-specific data

such that a possible increment or decrement in the performance of the model could be noticed in future studies.

6. Conclusion

This paper introduced the development and evaluation of a fine-tuned large language model that is specifically designed for legal tasks and queries in the Indian judicial system. The capability of this model was quite impressive, as it generated accurate responses to legal queries in various languages and obtained high similarity scores. Unlike other models—for instance, the ChatGPT model—this model has greatly reduced instances of hallucination, making the outputs reliable and trustworthy.

Of particular mention is the performance of this Legal Assist AI model on the AIBE examination, where the model scored an impressive 60% score, surpassing the performance of other models tested earlier. This demonstrates the potential of the model to assist legal practitioners by carrying out tedious tasks, such as looking for relevant sections and articles pertaining to a specific case, and thus contributing in the field.

Quality performance of the model can be enhanced by supplementing quality training data with more expert reviews and additional case specific data. However, it should be noted that increasing the data amount may result in either significant improvements or potential decreases in accuracy, respectively. Continuous improvement will ensure that the Legal Assist AI model remains a useful tool for the general public, providing them accurate and efficient assistance.

Declarations

Declaration of Generative AI and AI-assisted technologies in the writing process:

During the preparation of this work, the authors used Grammarly to correct the grammar across the entire content and to paraphrase sentences wherever deemed necessary. After using these tools and services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. Additionally, authors have presented test samples from the dataset that were fed into the model 'as such' in Table 3. Therefore, the questions and their corresponding answers have been put in quotes. The responses generated by the proposed LLM model (Legal Assist) have also been presented within the same Table in quotes.

Author Contribution

All authors contributed significantly to the research and preparation of this manuscript. A.I.A. conceptualized the study and supervised the research process. J.G. and A.S. were responsible for data collection and conducted the formal analysis. J.G. and S.S. carried out the literature review and drafted the initial manuscript. A.I.A. provided critical revisions and finalized the manuscript for submission. All authors read and approved the final version of the manuscript.

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Figures

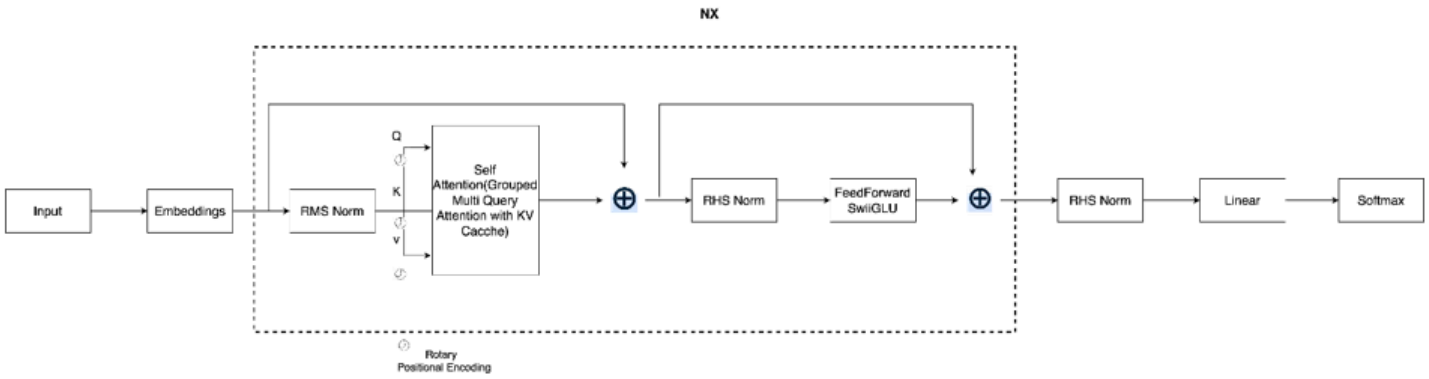


Figure 1

Transformer Model Architecture

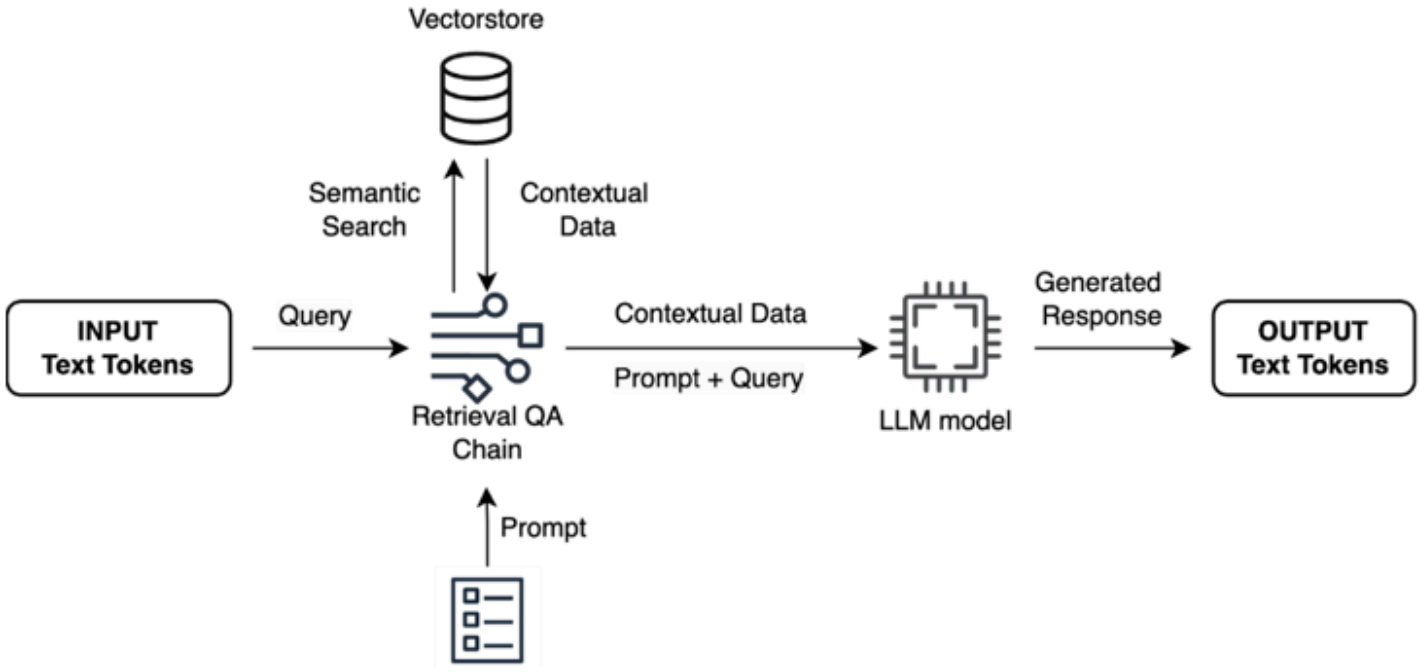


Figure 2

Proposed Legal Assist AI Architecture

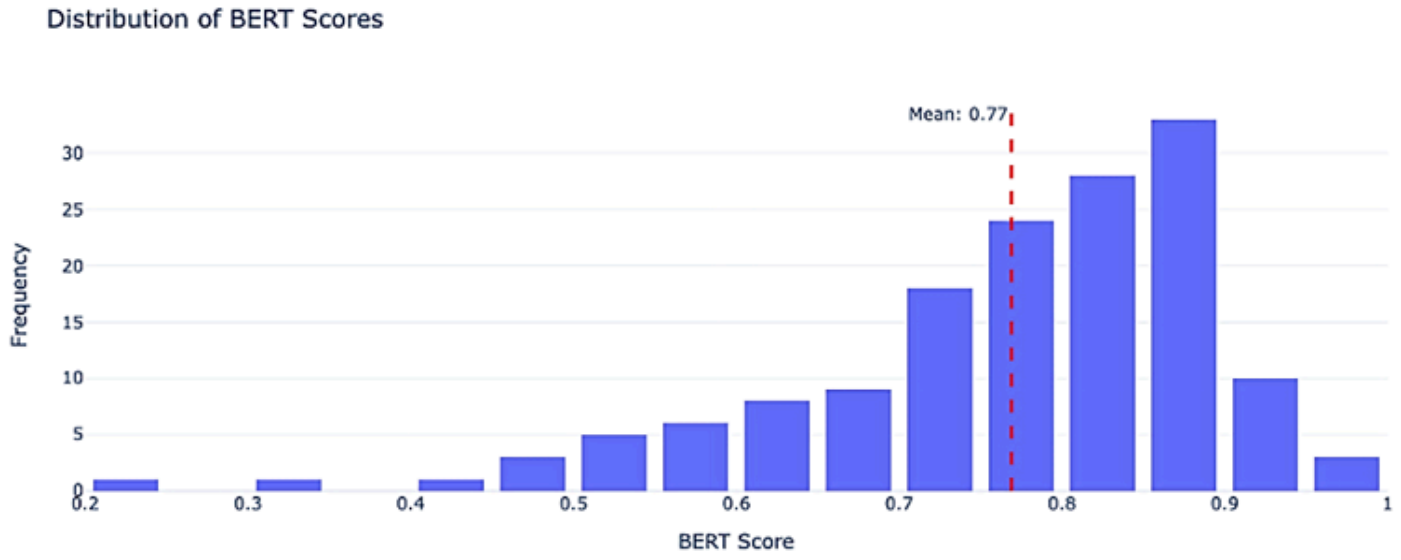


Figure 3

Distribution of BERT scores across different 150 questions